

Project Title
Word-Level Plagiarism Checker for Short Documents
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents a Word-Level Plagiarism Checker for detecting copied content in short documents such as essays, assignments, and reports. The system is developed using manually implemented data structures and algorithms without relying on prebuilt library implementations. Documents are tokenized into normalized word sequences and organized using a custom hash table and trie for efficient storage, indexing, and retrieval. Exact plagiarism detection is performed using the Rabin-Karp algorithm with polynomial rolling hash to identify matching words and phrases efficiently. To evaluate the overall similarity between documents, the system performs document similarity analysis using manually implemented Suffix Arrays and Longest Common Prefix (LCP) techniques, enabling the identification of common text segments and similarity scores. The system reports the corresponding source document and similarity percentage for each comparison. The project demonstrates the practical application of advanced string algorithms and custom data structures in building an efficient, reliable, and scalable plagiarism detection system.

Signature of the Guide